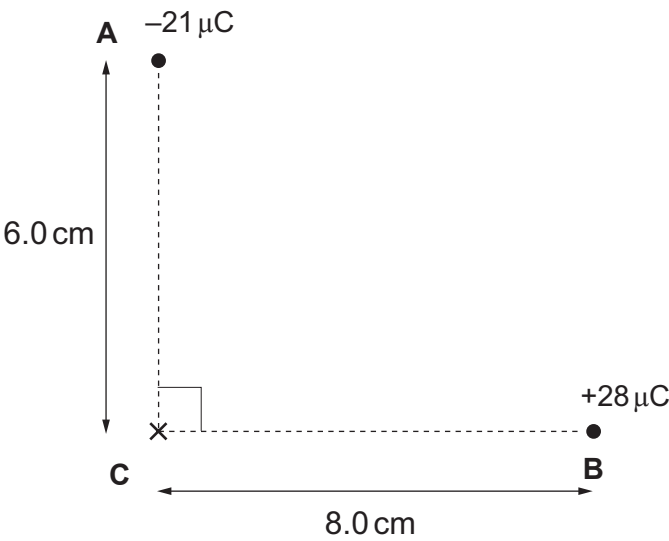


2. Two charges are placed at points **A** and **B** as shown in the diagram.



- (a) Draw **two** arrows on the diagram to represent the directions of the electric fields at **C** due to the $-21\,\mu\text{C}$ and $+28\,\mu\text{C}$ charges. [2]
- (b) Calculate the magnitude and direction of the resultant electric field at **C**. [5]

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(c) Show that the electrical potential at point **C** is zero. [2]

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(d) A positron is initially at rest at point **C** and accelerates rapidly due to the electric field. Calculate the potential at the point where the velocity of the positron is $5.4 \times 10^7 \text{ ms}^{-1}$. [3]

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